

CSC311H5 Introduction to Machine Learning

Lecture 10

University of Toronto Mississauga

Test #2 Reminders

- Friday Mar 20, 7:15-8:45pm, IB110/IB120
- Covers materials up to the end of week 8, with a focus on weeks 5-8
- See Quercus for the list of learning objectives tested
- Make sure to **bring your own non-programmable calculator**
 - ▶ calculators cannot be shared
- You may bring one single-sided cheat sheet

Cheat Sheet Rules

- Must be one single-sided sheet of paper (or no more than 4000 characters if you cannot use a sheet of paper for accessibility reasons).
- Must be created by and unique to the student who is using it. You may not copy another student's cheat sheet.
- Must have student's name and student number clearly at the top.
- Can be hand-written or typed. Whatever fits on one page is fine.
- Will be collected with the test. If you want to keep a copy, you must make your own copy in advance.

Review

Last week, we began the discussion on **probabilistic models** and **generative classifiers**

Discriminative classifiers: Model $p(c|\mathbf{x})$ directly

- Example: logistic regression

Generative classifiers: Model $p(\mathbf{x}|c)$, then Bayes Rule $p(c|\mathbf{x}) = \frac{p(\mathbf{x}|c)p(c)}{p(\mathbf{x})}$

- Example: Naive Bayes! Where we assume
 - ▶ $x_i \in 0, 1$
 - ▶ x_i are conditionally independent given c

$$p(\mathbf{x}, c) = p(c)p(x_1|c)p(x_2|c) \dots p(x_D|c)$$

Naive Bayes Log Likelihood, Parameters $\theta = \{\pi, \theta_{jc}\}$

$$\begin{aligned}\ell(\theta) &= \sum_{i=1}^N \log p(c^{(i)}, \mathbf{x}^{(i)}) = \sum_{i=1}^N \log \left\{ p(\mathbf{x}^{(i)} | c^{(i)}) p(c^{(i)}) \right\} \\&= \sum_{i=1}^N \log \left\{ p(c^{(i)}) \prod_{j=1}^D p(x_j^{(i)} | c^{(i)}) \right\} \\&= \sum_{i=1}^N \left[\log p(c^{(i)}) + \sum_{j=1}^D \log p(x_j^{(i)} | c^{(i)}) \right] \\&= \underbrace{\sum_{i=1}^N \log p(c^{(i)})}_{\text{Bernoulli log-likelihood of labels}} + \sum_{j=1}^D \underbrace{\sum_{i=1}^N \log p(x_j^{(i)} | c^{(i)})}_{\text{Bernoulli log-likelihood for feature } x_j}\end{aligned}$$

Estimating Parameters

- Maximum likelihood estimation (MLE)
 - ▶ Choose the parameters that maximize the (log) likelihood
 - ▶ Problem: Data sparsity
- Bayesian Inference
 - ▶ Treat the parameters as random variables with a **prior distribution**
 - ▶ Problem: Can be expensive
- MAP Estimate
 - ▶ Choose the parameters that maximize the posterior distribution
 - ▶ Solves the data sparsity problem

From Naive Bayes to GDA

- Naive Bayes works for binary features
 - ▶ Use the conditional independence assumption

$$p(x_1, \dots, x_D, c) = p(x_1|c) \dots p(x_D|c)$$

- ▶ Choose a simple (Bernoulli) distribution for each $p(x_j|c)$
- What about features $\mathbf{x}$ that are continuous?
 - ▶ Choose a simple (**Gaussian**) distribution for each $p(x_j|c)$

Next, we will discuss fitting **Gaussian distributions** to our data.

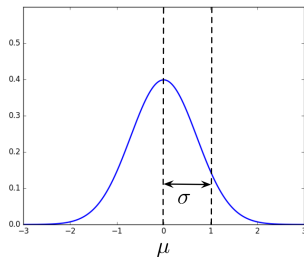
Section 1

Univariate Gaussian Discriminant Analysis

The (Univariate) Gaussian Distribution

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

- Parameterized by mean μ and variance σ^2 .
- The Central Limit Theorem says that sums of lots of independent random variables are approximately Gaussian.
- In machine learning, we use Gaussians a lot because they make the calculations easy.



Learning the Parameters of a Gaussian Distribution

If we have two possible classes and just one scalar input feature, we can write the likelihood of our data as:

$$\mathcal{L} = p(\mathcal{D}) = \prod_{i=1}^N p(x^{(i)}, c^{(i)}) = \prod_{i=1}^N p(x^{(i)} | c^{(i)}) p(c^{(i)})$$

We follow the same procedure as we have for MLE before, this time knowing that we will eventually substitute a Gaussian for $p(x^{(i)} | c^{(i)})$. Start by taking the log to get the **log-likelihood** $\ell = \log \mathcal{L}$:

$$\ell = \log p(\mathcal{D}) = \sum_{i=1}^N \left(\log p(x^{(i)} | c^{(i)}) + \log p(c^{(i)}) \right)$$

Learning the Parameters of a Gaussian Distribution

Split the sum by class, so that each term in the summation has the same distribution:

$$\ell = \sum_{i:c^{(i)}=0} \left(\log p(x^{(i)} | c^{(i)}) + \log p(c^{(i)}) \right) + \sum_{i:c^{(i)}=1} \left(\log p(x^{(i)} | c^{(i)}) + \log p(c^{(i)}) \right)$$

Then, the first sum can be written:

$$\sum_{i:c^{(i)}=0} \left(\log p(x^{(i)} | c = 0) + \log(1 - \pi) \right) = \sum_{i:c^{(i)}=0} \left(\log \mathcal{N}(x; \mu_0, \sigma_0^2) + \log(1 - \pi) \right)$$

And the second sum can be written:

$$\sum_{i:c^{(i)}=1} \left(\log p(x^{(i)} | c = 1) + \log \pi \right) = \sum_{i:c^{(i)}=1} \left(\log \mathcal{N}(x; \mu_1, \sigma_1^2) + \log \pi \right)$$

So, we have to learn the mean (μ) and variance (σ^2) of one Gaussian for each class, and π , which is Bernoulli. Since we have done MLE for Bernoulli distributions before, we will focus on the Gaussians here.

Learning the Parameters of a Gaussian Distribution

We optimize each Gaussian separately. For each, we get:

$$\begin{aligned}\ell &= \sum_{i:c^{(i)}=j} \left(\log \mathcal{N}(x; \mu_1, \sigma_1^2) \right) \\ &= \sum_{i:c^{(i)}=j}^N \left[-\frac{1}{2} \log(2\pi) - \log \sigma - \frac{(x^{(i)} - \mu)^2}{2\sigma^2} \right] \\ &= -\frac{N_j}{2} \log(2\pi) - N_j \log \sigma - \frac{1}{2\sigma^2} \sum_{i:c^{(i)}=j}^N (x^{(i)} - \mu)^2.\end{aligned}$$

(Note that π here is the constant π . The Bernoulli π does not appear here.)

Learning the Parameters of a Gaussian Distribution

Differentiate ℓ with respect to μ and set to zero:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i:c(i)=j}^N (x^{(i)} - \mu) = 0 \quad \Rightarrow \quad \sum_{i:c(i)=j}^N x^{(i)} = N_j \mu \quad \Rightarrow$$

$$\mu_{\text{MLE}} = \frac{1}{N_j} \sum_{i:c(i)=j}^N x^{(i)}.$$

So the maximum-likelihood estimate of the mean is the **sample mean** of the data.

Learning the Parameters of a Gaussian Distribution

Differentiate ℓ with respect to σ^2 (treat σ^2 as the variable) and set to zero.

Let $\tau = \sigma^2$. Then $\ell = -\frac{N}{2} \log(2\pi) - \frac{N}{2} \log \tau - \frac{1}{2\tau} \sum_i (x^{(i)} - \mu)^2$.

Setting $\frac{\partial \ell}{\partial \tau} = -\frac{N}{2\tau} + \frac{1}{2\tau^2} \sum_i (x^{(i)} - \mu)^2 = 0$ gives $\tau = \frac{1}{N} \sum_i (x^{(i)} - \mu)^2$.

Using the MLE for μ in that expression, we obtain:

$$\sigma_{\text{MLE}}^2 = \frac{1}{N_j} \sum_{i:c^{(i)}=j} (x^{(i)} - \mu_{\text{MLE}})^2.$$

So the maximum-likelihood estimate of the variance is the **sample variance** (average squared deviation from the sample mean).

Univariate GDA Exercise

We now have everything we need for 1D problems! To build a generative model, we fit $p(x|c = 0)$, $p(x|c = 1)$, and $p(c)$.

Then, use the Bayes Rule to make predictions:

$$p(c = 1|x) = \frac{p(x|c = 1)p(c = 1)}{p(x|c = 1)p(c = 1) + p(x|c = 0)p(c = 0)}$$

Exercise: Estimate the parameters μ_0 , σ_0^2 , μ_1 , σ_1^2 , and π using MLE for GDA on the table of data below.

x	c
5	0
7	0
9	0
10	0
9	1
11	1
13	1

Univariate GDA Exercise

Solution:

$$\mu_0 = \frac{1}{N} \sum_{i=0}^N x^{(i)} = \frac{7+9+10+5}{4} = \frac{31}{4} = 7.75$$

$$\sigma_0^2 = \frac{1}{N} \sum_{i=0}^N (x^{(i)} - \mu_0)^2 = \frac{1}{4} [(7 - 7.75)^2 + (9 - 7.75)^2 + (10 - 7.75)^2 + (5 - 7.75)^2] \approx 3.69$$

$$\mu_1 = \frac{1}{N} \sum_{i=1}^N x^{(i)} = \frac{9+11+13}{3} = \frac{33}{3} = 11$$

$$\sigma_1^2 = \frac{1}{N} \sum_{i=1}^N (x^{(i)} - \mu_1)^2 = \frac{1}{3} [(9 - 11)^2 + (11 - 11)^2 + (13 - 11)^2] \approx 2.67$$

$$\pi = \frac{N_1}{N} = \frac{3}{7} \approx 0.429$$

Univariate GDA Exercise

Exercise: Using this model, give the probability that a test data point with $x = 10$ belongs to class 1.

Univariate GDA Exercise

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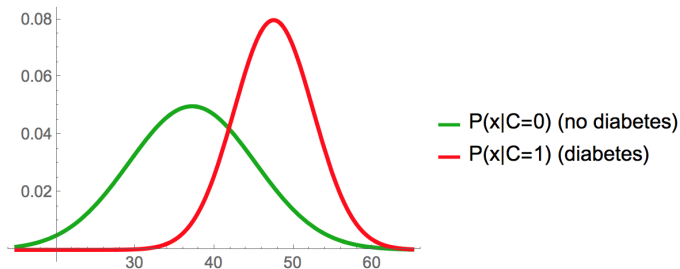
Solution:

$$P(c = 1|x = 10) \propto P(x = 10|c = 1)P(c = 1) = \mathcal{N}(10; \mu_1, \sigma_1^2)\pi \approx 0.2025(0.429) \approx 0.0868$$

$$P(c = 0|x = 10) \propto P(x = 10|c = 0)P(c = 0) = \mathcal{N}(10; \mu_0, \sigma_0^2)(1 - \pi) \approx 0.1046(0.571) \approx 0.0598$$

$$P(c = 1|x = 10) \approx \frac{0.0868}{0.0868+0.0598} \approx 0.592$$

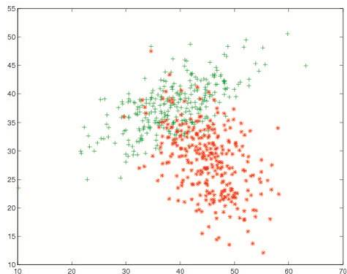
Diabetes Example (1D)



So far, we have been working with a univariate Gaussians for a single input feature.

Now, we are ready to consider multivariate Gaussians for a vector of input features.

Diabetes Example (multiple features $\mathbf{x}$)



- Fit **Multivariate Gaussians** $p(\mathbf{x}|c = 0)$ and $p(\mathbf{x}|c = 1)$
- Use the Bayes Rule to make predictions

$$p(c = 1|\mathbf{x}) = \frac{p(\mathbf{x}|c = 1)p(c = 1)}{p(\mathbf{x}|c = 1)p(c = 1) + p(\mathbf{x}|c = 0)p(c = 0)}$$

Gaussian Discriminant Analysis

- **Gaussian Discriminant Analysis** in its general form assumes that $p(\mathbf{x}|c)$ is distributed according to a multivariate Gaussian distribution
- Multivariate Gaussian distribution:

$$p(\mathbf{x}|c = k) = \frac{1}{(2\pi)^{D/2} |\mathbf{\Sigma}_k|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_k)^T \mathbf{\Sigma}_k^{-1} (\mathbf{x} - \boldsymbol{\mu}_k) \right]$$

where $|\mathbf{\Sigma}_k|$ denotes the determinant of the matrix.

- Each class k has associated mean vector $\boldsymbol{\mu}_k$ and covariance matrix $\mathbf{\Sigma}_k$
 - ▶ Each $\boldsymbol{\mu}_k$ has D parameters, for DK total.
 - ▶ Each $\mathbf{\Sigma}_k$ has $\mathcal{O}(D^2)$ parameters, for $\mathcal{O}(D^2K)$ — could be hard to estimate (more on that later).

GDA Learning

How do we fit the parameters μ_k and Σ_k in our model?

$$p(\mathbf{x}|c = k) = \frac{1}{(2\pi)^{D/2}|\Sigma_k|^{1/2}} \exp \left[-\frac{1}{2}(\mathbf{x} - \mu_k)^T \Sigma_k^{-1}(\mathbf{x} - \mu_k) \right]$$

- Maximum likelihood; Bayes Estimation like in Naive Bayes!
- In order to fit these parameters, we need to know a bit more about:
 - 1 multivariate Gaussian
 - 2 matrix decompositions

Section 2

Multivariate Gaussian

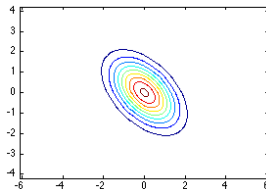
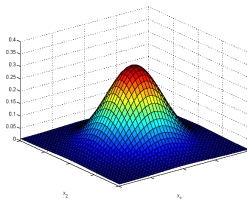
Multivariate Gaussian Probability Distribution Function

PDF of the univariate Gaussian distribution:

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

PDF of the multivariate Gaussian distribution:

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2} |\boldsymbol{\Sigma}|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right]$$



Mean and Covariance

- Mean

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{x}] = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_D \end{pmatrix}$$

- Covariance

$$\boldsymbol{\Sigma} = \text{Cov}(\mathbf{x}) = \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu})(\mathbf{x} - \boldsymbol{\mu})^\top] = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1D} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2D} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1D} & \sigma_{2D} & \cdots & \sigma_D^2 \end{pmatrix}$$

- The statistics ($\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$) uniquely define a **multivariate Gaussian** (or **multivariate Normal**) distribution, denoted $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ or $\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$

Bivariate Gaussian Examples

$$\Sigma = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Sigma = 0.5 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Sigma = 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

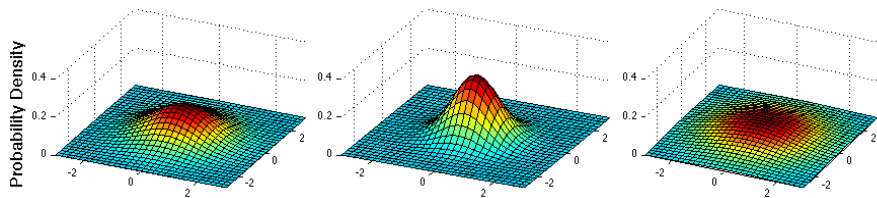


Figure 1: Probability density function

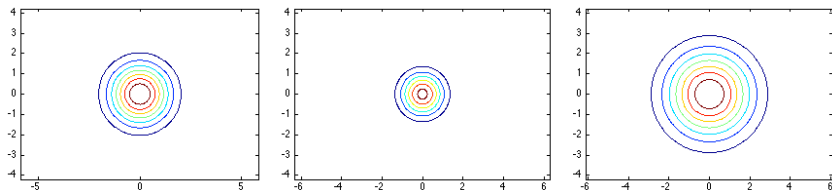
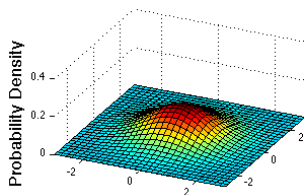


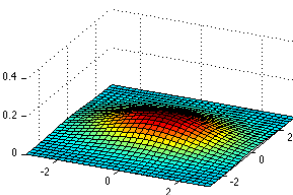
Figure 2: Contour plot of the pdf

Bivariate Gaussian Examples

$$\Sigma = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$



$$\Sigma = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$



$$\Sigma = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

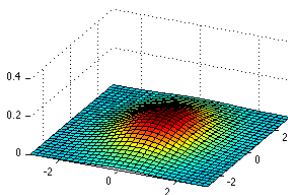


Figure 3: Probability density function

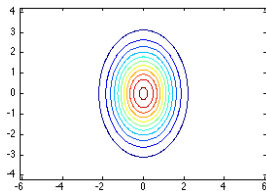
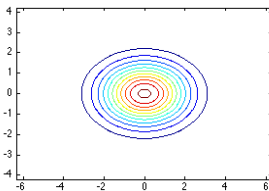
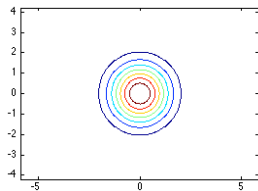


Figure 4: Contour plot of the pdf

Bivariate Gaussian Examples

$$\Sigma = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix} \quad \Sigma = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}$$

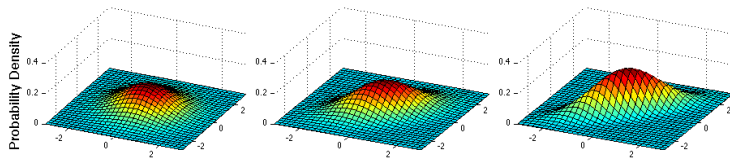


Figure 5: Probability density function

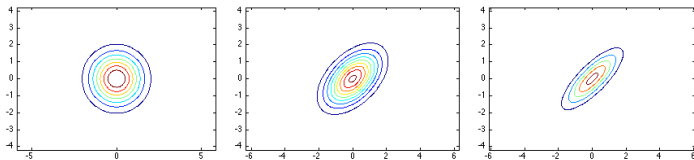
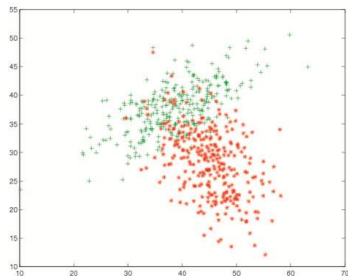


Figure 6: Contour plot of the pdf

Multivariate Gaussian MLE

Back to the diabetes example:



How can we fit a multivariate Gaussian for a single class via maximum likelihood?

Multivariate Gaussian MLE

Assume each $\mathbf{x}^{(i)}$ is drawn from a multivariate Gaussian distribution with mean $\boldsymbol{\mu}$, and covariance $\boldsymbol{\Sigma}$. Then we have the following log-likelihood function:

$$\begin{aligned}\ell(\boldsymbol{\mu}, \boldsymbol{\Sigma}) &= \log \prod_{i=1}^N \left[\frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x}^{(i)} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}^{(i)} - \boldsymbol{\mu}) \right\} \right] \\ &= \sum_{i=1}^N \log \left[\frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x}^{(i)} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}^{(i)} - \boldsymbol{\mu}) \right\} \right] \\ &= \sum_{i=1}^N \underbrace{-\log(2\pi)^{d/2}}_{\text{constant}} - \log |\boldsymbol{\Sigma}|^{1/2} - \frac{1}{2} (\mathbf{x}^{(i)} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}^{(i)} - \boldsymbol{\mu})\end{aligned}$$

Gaussian Maximum Likelihood: Mean

Maximize the log-likelihood by setting the derivative to zero:

$$\frac{\partial \ell}{\partial \mu_j} = -\frac{\partial}{\partial \mu_j} \sum_{i=1}^N \frac{1}{2} (\mathbf{x}^{(i)} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x}^{(i)} - \boldsymbol{\mu}) = 0$$

$$\nabla_{\boldsymbol{\mu}} \ell = -\sum_{i=1}^N \boldsymbol{\Sigma}^{-1} (\mathbf{x}^{(i)} - \boldsymbol{\mu}) = 0$$

- Here we use the identity $\nabla_{\mathbf{x}} \mathbf{x}^T A \mathbf{x} = 2A\mathbf{x}$
- Solving we get $\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}^{(i)}$.
- This is just the sample mean of the observed values, or the **empirical mean**.

Gaussian Maximum Likelihood: Covariance

- We can do a similar calculation for the covariance matrix Σ (we skip the details).
- Setting the partial derivatives to zero, just like before, we get:

$$\begin{aligned} 0 = \nabla_{\Sigma} \ell &\implies \hat{\Sigma} = \frac{1}{N} \sum_{i=1}^N (\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}})(\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}})^{\top} \\ &= \frac{1}{N} (\mathbf{X} - \mathbb{1}\hat{\boldsymbol{\mu}}^{\top})^{\top} (\mathbf{X} - \mathbb{1}\hat{\boldsymbol{\mu}}^{\top}) \end{aligned}$$

where $\mathbb{1}$ is an N -dimensional vector of 1s.

- This is called the **empirical covariance** and comes up quite often (e.g., PCA soon!)
- Derivation in multivariate case is tedious. No need to worry about it. But it is good practice to derive the MLE estimate of σ one dimension. See math review.

Gaussian Maximum Likelihood: Summary

We can compute the ML estimates of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ in closed form:

$$\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}^{(i)}$$

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{N} \sum_{i=1}^N (\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}})(\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}})^{\top}$$

Gaussian Maximum Likelihood: Summary

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$$\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}^{(i)}$$
$$\hat{\boldsymbol{\Sigma}} = \frac{1}{N} \sum_{i=1}^N (\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}})(\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}})^{\top}$$

To understand $\boldsymbol{\Sigma}$ and its effect on the Gaussian PDF, we need to understand the spectral decomposition of a matrix.